

Academic Review Report

Multiplicative GARCH-X with VIX: Daily Source Decomposition Beats GARCH-MIDAS for Volatility Forecasting and Risk Management

Document type: Journal paper (target: J. Empirical Finance / J. Forecasting)

Review date: 2026-04-10

Review standard: Strict (referee perspective)

Review Summary

HIGH — must fix before submission	7 items
MEDIUM — recommended fix	12 items
LOW — optional improvement	8 items
Overall assessment	Good — publishable after revisions

General impression. The paper makes a clear and well-motivated argument that a parsimonious daily multiplicative GARCH-X model with VIX dominates the more complex GARCH-MIDAS framework when the external variable is daily. The 17-specification horse race is systematic, the DM test statistics are strong (especially A4f vs. GJR at $t = 4.48$), the VRP interpretation adds intellectual value, and the honest null result on VRP predictability is commendable. However, several issues—ranging from numerical inconsistencies between the paper and experiment data to selective reporting in the VRP section, incomplete citations, and methodological omissions—need to be addressed before submission.

I. Logic Structure

Strengths:

- The Introduction clearly states the gap (VIX is daily, so MIDAS mixed-frequency aggregation is conceptually redundant) and previews three contributions with key numbers.
- The progression from specification horse race → economic interpretation (VRP) → risk management application → cross-asset validation is logical and natural.
- The “source decomposition” label is well motivated and clearly distinguished from the long-run/short-run narrative of GARCH-MIDAS.

Issue M1 (MEDIUM): Missing Discussion section. The paper jumps from Robustness (Section 6) directly to Conclusion (Section 7), but the JBF-standard structure calls for a Discussion section that provides economic interpretation, connects findings to theory, and discusses caveats. Currently, economic interpretation is scattered across Sections 5.1–5.3 and the Conclusion. A dedicated Discussion section would strengthen the narrative.

Issue M2 (MEDIUM): Robustness section is thin. Section 6 contains only four short subsections (refit frequency, estimation consistency, multiple testing, Student- t degrees of freedom). Missing robustness analyses include:

- Sub-period analysis (e.g., pre-COVID 2019–2020 vs. post-COVID 2021–2026).
- Alternative OOS start dates.
- Sensitivity to estimation window size ($W = 1500, 2000, 2500$).

- Alternative volatility proxies (5-minute RV instead of r_t^2).

At least the sub-period analysis should be included, as a referee will want to know whether the A4f dominance holds in both calm and turbulent sub-periods.

II. Claim-Evidence Matching

Issue H1 (HIGH): MIDAS-FS DM test statistics in Table 3 do not match experiment data. Cross-checking against the K988b experiment results JSON reveals the following discrepancies:

Model	Paper DM t	Data DM t	Difference	Paper Harvey sig.
C1 (MIDAS-FS, $K_m = 6$)	3.55	2.85	+0.70	Yes $\rightarrow$ should be No
C2 (MIDAS-FS, $K_m = 12$)	2.33	1.71	+0.62	No (correct)
C3 (MIDAS-FS, $K_m = 24$)	2.85	2.19	+0.66	No (correct)

The most critical error is C1: the paper claims $t = 3.55$ (Harvey-significant), but the experiment data shows $t = 2.85$ (not significant at $|t| > 3.0$). This changes the finding count from “14 of 16 models achieve $|t| > 3.0$ ” (stated in Section 6.3) to 13 of 16. Additionally, the K988b GJR baseline QLIKE (-8.273) differs slightly from the K988 baseline (-8.277), which may account for some discrepancy—but not the ~ 0.7 -unit gap in DM t .

Fix: Re-verify all MIDAS-FS DM statistics using a single consistent GJR baseline. Update Table 3, the abstract (“14 of 16” claim in Section 6.3), and any downstream text.

Issue H2 (HIGH): Selective reporting of VRP predictive regression results. The paper states “the Newey-West t -statistic on g_t ranges from -1.28 to -1.95 across horizons—none surpassing the Harvey threshold.” However, this range refers only to models 1–2 (g-only and VRP+g). The experiment data (K998) shows that models 3–4 (with additional controls: log-VIX, full controls) produce t -statistics as large as -2.34 ($h = 5$, model3) and -2.27 ($h = 1$, model3). While none exceed the Harvey $|t| > 3.0$ threshold, the selective reporting of only the weakest specifications is misleading.

Fix: Either report the full range (“from -1.28 to -2.34 ”) or present a table with all model specifications and horizons. The conclusion that g_t has no exploitable predictive power remains valid, but transparency requires reporting the strongest evidence too.

Issue M3 (MEDIUM): Cross-asset table mixes experiment sources without disclosure. Table 5 (cross-asset validation) reports GLD with GVZ ($t = -3.39$), but this value comes from the dual-factor VIX+GVZ model in K997 (Table 6 experiment), not from the standard A4f specification in K994. Table 5 presents it as “A4f” which is misleading—it is actually a dual-factor $\tau_t = \theta_0 + \theta_1 \text{VIX}^2 + \theta_2 \text{GVZ}^2$ model with 3 τ -parameters vs. 2 for the standard A4f.

Fix: Either (a) report the single-factor GVZ result from K997 ($t = -3.17$) in Table 5, which is still Harvey-significant, or (b) clearly label the GLD entry as “A4f-GVZ” or “A4f-dual” and note the model differs from the other rows. Option (a) is cleaner.

Issue M4 (MEDIUM): Abstract claims “scorecard 3/4” but the 2.5% VaR fails. The abstract and conclusion state A4f- t achieves “VaR backtesting at two of three confidence levels (scorecard 3/4).” The VaR 2.5% test has DQ $p = 0.006$ (clear failure), and the scorecard counts 2 VaR passes + 1 ES pass = 3/4. This is technically correct but the scorecard construction is unusual: treating VaR at 3 levels + ES at 1 level as a “4-test scorecard” is not standard. A referee might argue this cherry-picks the denominator.

Fix: Explicitly acknowledge in the text that the 2.5% level fails (DQ $p = 0.006$), and discuss why—is it the DQ test’s sensitivity to violation clustering? Consider presenting it as “passes 2 of 3 VaR levels and 1 of 1 ES level” rather than conflating them into a single scorecard.

III. Mechanical vs. Empirical Distinction

Strengths:

- The paper correctly notes that VIX-squared dominance follows from dimensional consistency—this is acknowledged as a dimensional argument, not an empirical discovery.
- The null result on VRP predictability is an honest empirical finding.

Issue M5 (MEDIUM): “Source decomposition” risks being perceived as tautological. If $\tau_t \approx \theta_0 + \theta_1 \text{VIX}^2$ and $g_t = \sigma_t^2 / \tau_t$, then $g_t \approx \sigma_t^2 / (\theta_0 + \theta_1 \text{VIX}^2)$. The correlation between g_t and VRP ($\rho \approx 0.80$) follows mechanically from the model specification, not from an independent empirical test. The paper should more explicitly address the anti-tautology test: what *independent* evidence shows that the GARCH dynamics in g_t add information beyond the mechanical ratio?

The comparison with the raw ratio ($\rho = 0.15$) partially addresses this, but the argument could be strengthened by decomposing the $\rho = 0.80$ into what comes from the ratio structure and what comes from the GARCH smoothing.

Fix: Add a brief discussion (2–3 sentences) acknowledging that the high ρ is partly mechanical, then arguing that the GARCH autoregressive structure is the empirical contribution (e.g., show that an AR(1) filter on the raw ratio achieves an intermediate ρ , demonstrating that the GARCH dynamics specifically add value).

IV. Statistical Rigor

Issue H3 (HIGH): No Model Confidence Set (MCS) test. The paper cites Hansen et al. (2011) in the bibliography but never uses the MCS. For a 17-model horse race, the MCS is a standard tool that identifies the set of models that are statistically indistinguishable from the best. A referee at J. Empirical Finance will almost certainly request this. Given that the top 10 QLIKE values span only -8.361 to -8.348 (a range of 0.013), it is likely that many models fall within the MCS, which would *qualify* the “A4f dominates” claim.

Fix: Run the MCS at $\alpha = 0.10$ and $\alpha = 0.25$. Report which models survive. If A4f is the sole survivor, this strengthens the paper enormously. If 5–10 models survive, the message shifts to “A4f is among the best, and daily GARCH-X models dominate MIDAS-FS” (still a strong finding).

Issue H4 (HIGH): Pairwise DM tests insufficient for a 17-model comparison. The paper conducts 16 pairwise DM tests (each model vs. GJR), but does not test any GARCH-X model against another (e.g., A4f vs. B1, A4f vs. A2). This means the claim “uniformly dominating all GARCH-MIDAS specifications” (abstract, introduction, conclusion) is based on QLIKE rank ordering, not statistical tests. The DM test between A4f (QLIKE = -8.361) and B1 (QLIKE = -8.354) may not be significant.

Fix: Add pairwise DM tests between A4f and each GARCH-MIDAS specification (B1–B3, C1–C3). If A4f vs. B1 is not significant, soften “uniformly dominating” to “outperforming in point estimates.”

Issue M6 (MEDIUM): Harvey (2016) threshold applied inconsistently. The paper uses $|t| > 3.0$ as the significance threshold (following Harvey 2016 for multiple testing). However, Harvey (2016) is about controlling false discovery rates in factor zoo studies. For a pre-specified model comparison (A4f vs. GJR), the standard 5% DM critical value ($|t| > 1.96$) is appropriate, and the Bonferroni correction computed in Section 6.3 ($z \approx 2.95$) is the correct multiple-testing adjustment. Using $|t| > 3.0$ as a blanket threshold conflates these two justifications.

Fix: Clarify that $|t| > 3.0$ is adopted as a conservative threshold for the full 16-comparison horse race (close to the Bonferroni value of 2.95), and that for the primary A4f vs. GJR comparison, the standard $|t| > 1.96$ applies and is comfortably exceeded.

Issue M7 (MEDIUM): No Giacomini-White (2006) conditional test. The DM test assumes constant loss differential. In a rolling-window scheme with regime changes (COVID, 2022 draw-down, 2025 trade war), the loss differential may be time-varying. The Giacomini-White conditional predictive ability test would address this.

Fix: Add GW tests for at least A4f vs. GJR and A4f vs. B1. This is standard for J. Empirical Finance.

Issue M8 (MEDIUM): OOS R^2 interpretation for VRP. The paper reports OOS $R^2 = 1.3\%$ for g_t -based VRP forecasts. This is actually *positive*, which typically indicates some predictive content in the Campbell-Thompson framework. The paper dismisses it as evidence of overfitting (“negative at longer horizons”), but the $h = 1$ result deserves more nuanced discussion. Is 1.3% economically meaningful? Is it statistically significant (Clark-West test)?

Fix: Report the Clark-West p-value for $R_{\text{OOS}}^2 = 1.3\%$. The current K998 data shows CW $t = 0.24$, $p = 0.40$ —not significant. Stating this directly would be cleaner than calling it “overfitting.”

V. Equations and Symbols

Issue H5 (HIGH): Ambiguous subscript t^* in Equation (6). The standardized return is defined as $u_{t-1} = r_{t-1}/\sqrt{\tau_{t^*}}$ with “ $t^* \in \{t, t-1\}$ depending on whether the current or lagged τ is used for normalization.” This notation is nonstandard and confusing. The paper should define two variants explicitly:

$$\begin{aligned} u_{t-1}^{(a)} &= r_{t-1}/\sqrt{\tau_t} && \text{(contemporaneous),} \\ u_{t-1}^{(b)} &= r_{t-1}/\sqrt{\tau_{t-1}} && \text{(lagged).} \end{aligned}$$

Fix: Replace the t^* notation with explicit variant definitions.

Issue M9 (MEDIUM): VIX units not converted in τ_t . Equation (4) writes $\tau_t = \theta_0 + \theta_1 \text{VIX}_{t-1}^2$. However, VIX is quoted as annualized percentage volatility (e.g., $\text{VIX} = 20$ means 20%). $\text{VIX}^2 = 400$ in percentage-squared units. For τ_t to match σ_t^2 (which is in decimal daily variance units), the VIX must be converted: $\text{VIX}^2/(100^2 \times 252)$. The paper should clarify whether θ_1 absorbs this scaling or whether VIX is pre-converted.

Fix: Add a sentence clarifying the scaling convention. E.g., “We use VIX_{t-1} in its raw form (annualized percentage); the coefficients θ_0, θ_1 absorb the necessary unit conversions.”

Issue M10 (MEDIUM): ES formula sign convention. Equation (10) writes $\text{ES}_{\alpha,t}^{\text{Normal}} = \hat{\sigma}_t \left(-\frac{\phi(z_\alpha)}{\alpha} \right)$. This gives a positive number when $z_\alpha < 0$ (left tail). For VaR and ES to represent losses (negative returns), both should be negative or both should be positive. The paper should clarify the sign convention. If $\text{VaR}_\alpha < 0$ (a negative return threshold), then ES should also be negative.

Fix: Define the sign convention explicitly. Common practice: $\text{VaR}_\alpha = -\hat{\sigma}_t z_\alpha > 0$ (positive loss), or $\text{VaR}_\alpha = \hat{\sigma}_t z_\alpha < 0$ (threshold return). Be consistent.

VI. Citations

Issue H6 (HIGH): Three uncited references in bibliography. The following entries appear in the bibliography but are never cited in the text:

- **Bollerslev (1986)** — foundational GARCH paper. Must be cited (e.g., when first mentioning GARCH).
- **Glosten, Jagannathan, and Runkle (1993)** — the GJR-GARCH paper. The model is called “GJR” throughout but never cited. Must be cited at first mention of the GJR process.
- **Diebold and Yilmaz (2012)** — a spillover paper with no obvious connection to this manuscript. Should be removed unless cited.

Fix: Add `\citet{bollerslev1986}` where GARCH is first introduced (Section 1 or 3). Add `\citet{glosten1993}` where GJR is first mentioned (Equation 5 or earlier). Remove `\bibitem{diebold2012}` if not cited.

Issue M11 (MEDIUM): Missing key citations.

- **Engle (1982)** — ARCH original paper. Not cited anywhere. Should appear alongside Bollerslev (1986).
- **Nelson (1991)** — EGARCH, relevant when discussing asymmetric volatility models.
- **Diebold and Mariano (1995/2002)** — the DM test paper itself is never cited, only used implicitly. Should be cited at first mention of the DM test.
- **Giacomini and White (2006)** — conditional predictive ability, relevant for rolling-window evaluation (even if not used).
- **Wang and Ghysels (2015)** — the bibliography labels this as wang2017 but the publication year is 2015. Fix the citation key to avoid confusion.

Issue M12 (MEDIUM): Self-citation to unpublished work. The paper cites “Lai (2026). Volatility targeting as drawdown insurance. Working paper.” This citation is used only for the 0050.TW split-adjustment procedure, which is a minor data-cleaning step. A referee may question citing an unpublished working paper for a data procedure. Consider either describing the procedure directly or citing a published source.

VII. Tables

Strengths:

- Tables use booktabs style (no vertical lines, proper `\toprule`/`\midrule`/`\bottomrule`).
- Captions are mostly self-contained with table notes explaining acronyms and test details.

Issue M13 (MEDIUM): Table 1 has 10 GARCH-X specs but the abstract says “seven daily GARCH-X variants.” Table 1 lists $A1-A5 + A2f + A4f + A3f + A2n + A4n = 10$ models. The abstract counts only 7. The discrepancy arises because $A2n$, $A4n$, $A3f$ were added in K988b. Either update the abstract to “ten” or justify the count of seven (perhaps by excluding the normalized variants as sub-variants).

LOW Issue L1: Table 3 ranking inconsistency. $A2n$ (rank 9, $QLIKE = -8.350$) is listed *after* $A4n$ (rank 8, $QLIKE = -8.348$), but $A2n$ has the higher (better) $QLIKE$. This appears to be a ranking error. Verify and re-sort.

LOW Issue L2: Table 2 caption should clarify annualization. The table reports “Mean (ann.)” for both SPY returns and VIX, but VIX is natively annualized while SPY mean is computed from daily and annualized. Add a note clarifying the annualization method (e.g., $\times 252$ for mean, $\times \sqrt{252}$ for std).

LOW Issue L3: Table 2 missing daily statistics. Including daily mean, daily std, and autocorrelation coefficients (ρ_1, ρ_5 of r_t^2) would strengthen the data description and provide evidence of volatility clustering.

VIII. Academic Tone

Strengths:

- The paper generally maintains an academic tone and avoids practitioner language.
- Claims are hedged appropriately (“preliminary evidence,” “consistent with the interpretation”).

LOW Issue L4: Minor practitioner language. Section 5.4 mentions “a practically relevant finding for risk officers and regulators.” While acceptable in J. Empirical Finance, consider softening to “These results have implications for risk management practice.”

LOW Issue L5: “Beats” in the title. The word “beats” is informal. Consider “Outperforms” or “Dominates” for a journal title.

IX. Contribution Count and Depth

The paper has exactly 3 contributions—ideal for JBF/JEF.

However, the depth of Contribution 2 (source decomposition / VRP) is mixed:

- The VRP tracking result ($\rho \approx 0.80$) is striking but partly mechanical (see Issue M5).
- The null result on VRP predictability is honest and valuable, but the analysis is somewhat thin (4 bullet points without a formal table).

Issue M14 (MEDIUM): VRP analysis deserves a formal table. The four VRP predictability approaches (Granger, predictive regressions, OOS R^2 , variance swap) are listed as bullet points. A formal table with rows = horizon $\times$ model and columns = test statistic / p -value / significance would be much stronger, especially given that the full data (K998) contains 16 regression results (4 models $\times$ 4 horizons).

X. Limitations

Strengths:

- The Conclusion section explicitly discusses limitations: 5-asset sample, DQ failure at 2.5%, and cross-asset evidence for free omega.
- The null result on VRP predictability is reported honestly.

Issue H7 (HIGH): Missing limitation: r_t^2 as volatility proxy. The paper uses r_t^2 as the volatility proxy for QLIKE evaluation, citing Patton (2011) for robustness. However, Patton (2011) shows robustness of *rankings*, not absolute QLIKE values. The paper should acknowledge that:

1. Using 5-minute realized variance as the proxy might change the *magnitude* of QLIKE differences and DM statistics.
2. The VIX-based model might look relatively better or worse with RV as the target, because VIX and RV have a known non-trivial relationship.

Ideally, a robustness table using 5-min RV (which is available for SPY in the research system's data) should be included.

LOW Issue L6: Missing limitation: single-asset primary analysis. The main 17-model horse race is conducted only on SPY. While cross-asset validation covers 4 additional assets, only the A4f and A4 specifications are tested cross-asset (not the full 17). This limits the generalizability of the functional form ranking.

LOW Issue L7: No out-of-sample start date sensitivity. The OOS period begins January 2, 2019, which is fixed. Rolling the start date by ± 1 year would test whether the results are sensitive to the specific OOS window.

LOW Issue L8: Forecasting horizon limited to 1 day. All evaluations are one-step-ahead. Multi-step forecasts ($h = 5, 10, 22$) would be relevant for portfolio construction and risk management applications.

Detailed Issue List

HIGH — Must Fix

- H1. **Table 3: MIDAS-FS DM statistics do not match experiment data** (Section 5.1, Table 3). C1 paper $t = 3.55$, data $t = 2.85$; C2 paper $t = 2.33$, data $t = 1.71$; C3 paper $t = 2.85$, data $t = 2.19$. Re-verify against consistent GJR baseline.
- H2. **Selective reporting of VRP predictive regression t -statistics** (Section 5.2.2). Paper reports range -1.28 to -1.95 (models 1–2 only); data shows -1.28 to -2.34 (all 4 models). Report the full range.
- H3. **No Model Confidence Set (MCS)** (Section 5.1). For a 17-model comparison, MCS is expected by referees at J. Empirical Finance.
- H4. **No pairwise DM tests among top models** (Section 5.1). “Uniformly dominates” claim requires A4f vs. B1 (and other MIDAS) DM tests, not just QLIKE ordering.
- H5. **Ambiguous t^* subscript in Equation (6)** (Section 3.1.2). Replace with explicit variant definitions.
- H6. **Three uncited bibliography entries** (Bibliography). Bollerslev (1986), Glosten et al. (1993) must be cited; Diebold & Yilmaz (2012) should be removed.
- H7. **Missing limitation on r_t^2 as volatility proxy** (Section 7). Acknowledge and ideally provide 5-min RV robustness check.

MEDIUM — Recommended Fix

- M1. **Missing Discussion section** — add between Robustness and Conclusion.
- M2. **Thin Robustness section** — add sub-period analysis and window-size sensitivity.

- M3. **Cross-asset table mixes experiment sources** (Table 5) — GLD uses dual-factor VIX+GVZ from K997, not standard A4f.
- M4. **Scorecard 3/4 conflates VaR and ES tests** — present VaR and ES results separately.
- M5. **Source decomposition partly tautological** — add anti-tautology discussion.
- M6. **Harvey (2016) threshold conflated with Bonferroni** — clarify the justification for $|t| > 3.0$.
- M7. **No Giacomini-White conditional test** — add for A4f vs. GJR.
- M8. **OOS R^2 interpretation** — report Clark-West p -value instead of calling it “overfitting.”
- M9. **VIX units scaling not clarified** in Equation (4).
- M10. **ES sign convention ambiguous** in Equation (10).
- M11. **Missing citations:** Engle (1982), Nelson (1991), Diebold-Mariano (1995). Wang-Ghysels key/year mismatch.
- M12. **Self-citation to unpublished working paper** for 0050.TW split adjustment.
- M13. **Abstract says “seven” GARCH-X variants but Table 1 has ten.**
- M14. **VRP predictability analysis lacks formal table.**

LOW — Optional Improvement

- L1. Table 3: A2n/A4n ranking swap (A2n QLIKE $>$ A4n but ranked lower).
- L2. Table 2: clarify annualization method for mean and std.
- L3. Table 2: add daily statistics (ρ_1, ρ_5 of r_t^2).
- L4. Minor practitioner language in Section 5.4.
- L5. Title: “Beats” is informal — consider “Outperforms.”
- L6. Missing limitation: single-asset primary analysis.
- L7. Missing limitation: no OOS start date sensitivity.
- L8. Missing limitation: 1-day horizon only.

Summary and Recommendations

Overall assessment: The paper presents a solid empirical contribution with a clear message (daily GARCH-X with VIX beats GARCH-MIDAS). The main DM result ($t = 4.48$) is strong and survives Harvey thresholds. The source decomposition interpretation and the honest VRP null result add intellectual depth. However, the HIGH-priority issues—particularly the numerical discrepancies in Table 3, the missing MCS test, and the selective VRP reporting—must be addressed before submission.

Revision priority:

1. Fix the MIDAS-FS DM statistics (H1) — credibility issue.
2. Add MCS test (H3) and pairwise DM among top models (H4) — referees will request these.

3. Fix citation issues (H6) — easy but important.
4. Address VRP selective reporting (H2) and proxy limitation (H7).
5. Clarify equation notation (H5).
6. Add sub-period robustness, formal VRP table, GW test (M2, M7, M14).
7. Polish: Discussion section, cross-asset table clarity, scorecard presentation.

Target journal fit:

- **J. Empirical Finance:** Good fit. The paper is empirical, uses proper evaluation tools, and the 17-model comparison is the kind of systematic study JEF publishes. Adding the MCS and GW tests would seal the fit.
- **J. Forecasting:** Also a good fit, especially for the forecasting evaluation component. The VRP interpretation might be seen as tangential at this journal.
- **J. Banking and Finance:** Possible but would require deeper economic interpretation (the Discussion section suggested in M1 would help).